

Causal inference & studies

A quick visual reference for asking whether X actually causes Y

Causal inference

Asking whether an exposure actually produces an outcome, not just travels alongside it.

Study design

The architecture that determines how convincingly a causal claim can be supported.

Four things that separate cause from coincidence



Temporality

Cause must precede effect in time



Confounding

A third variable explaining both



Randomization

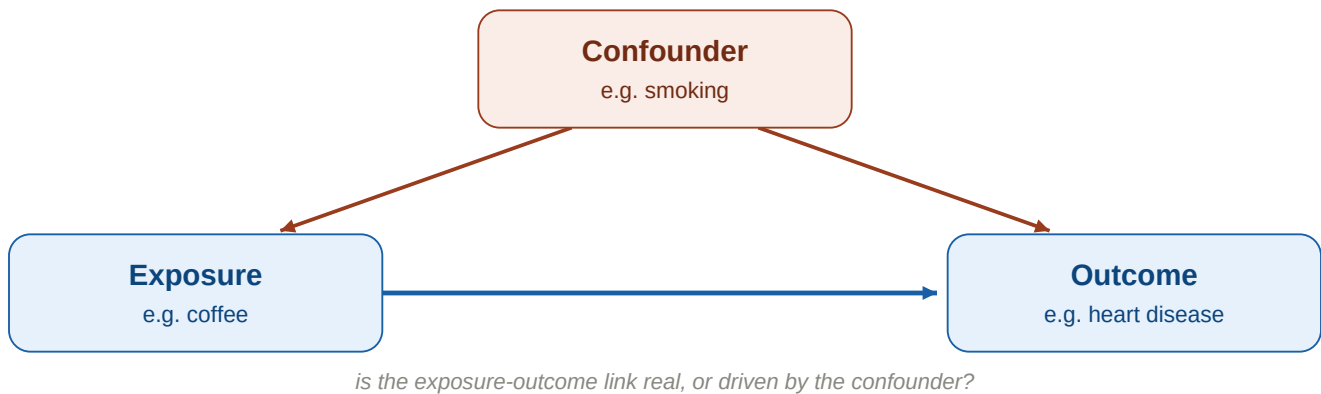
Breaks the link between exposure and confounders



Reverse causation

The outcome may be causing the exposure

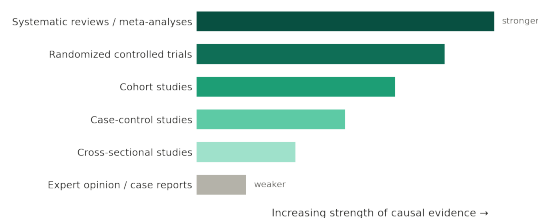
The confounding triangle (a causal diagram)



Bradford Hill criteria: features that strengthen a causal claim

- Strength
- Dose-response
- Consistency
- Plausibility
- Specificity
- Coherence
- Temporality
- Reversibility

Hierarchy of evidence across study designs



Quick reference: randomization is the strongest tool against confounding; observational designs need extra checks like Bradford Hill.